

Training Course Prometheus & Grafana



Course Overview



Module 1: Prometheus

Module 2: Grafana

Module 1: Prometheus Introduction



Why monitor?

- Analyzing long-term trends
- Comparing over time or experiment groups
- Alerts
- Dashboards
- Conducting ad-hoc retrospective analysis

*To answer the question: **What is broken and broken ?***

What to monitor ?

- ✓ **Host:** CPU, Memory, I/O, Network, Filesystem
- ✓ **Containers:** CPU, Memory, I/O, Network, Restart, Throttling
- ✓ **Applications:** Throughput, Latency

The Four Golden Signals

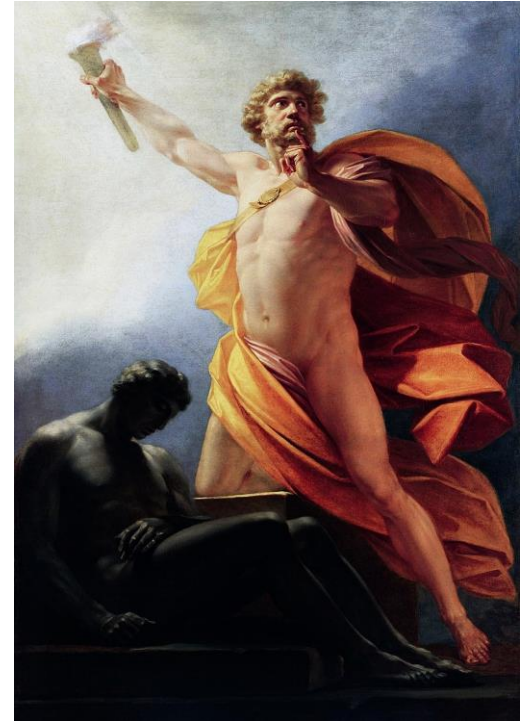
- Latency
- Traffic
- Error
- Saturation

Module 1: Prometheus Introduction

Prometheus?



Prometheus was presented to be the protector and benefactor of mankind.



Prometheus?



Prometheus's general info:

- Heavily inspired by Borgmon
- Built by ex-Googlers at SoundCloud
- Pull-based (scrapes at regular intervals)
- Many integrations possibilities

Prometheus is:

- Monitoring system and timeseries database
- Instrumentation
- Metrics collections and storage
- Querying
- Alerting
- Dashboard / Graphing / Trending

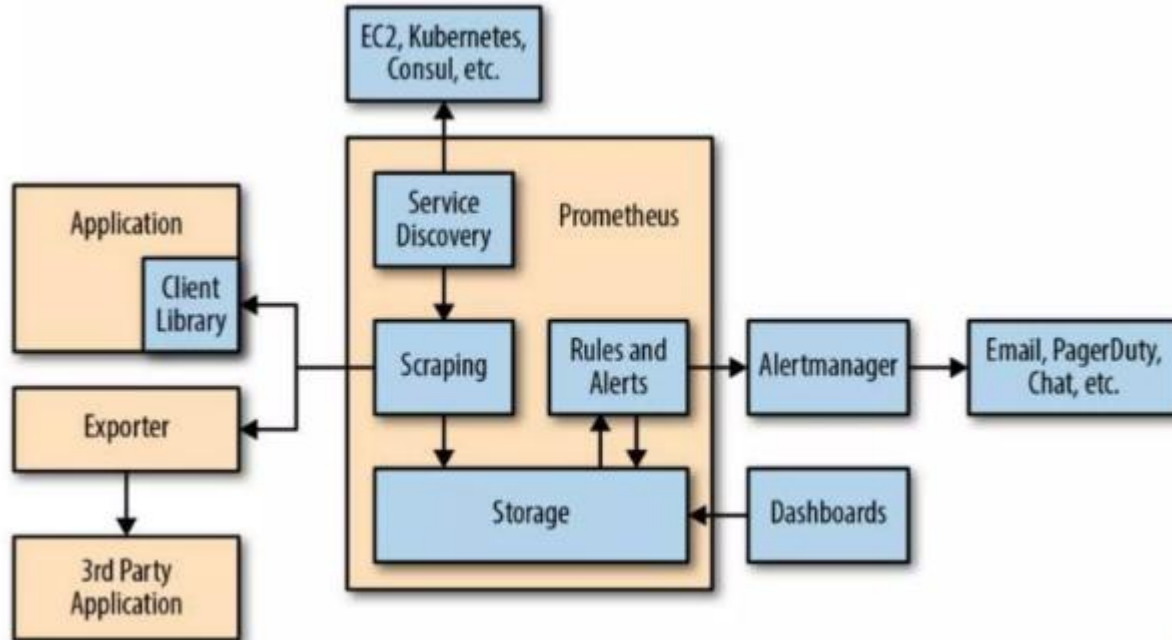
Prometheus focus on:

- Operational system monitoring
- Dynamic cloud environments

Prometheus does not do:

- Raw log / event collections (use ELK stack)
- Request tracing (use Opentracing.io)
- “Magic” anomaly detection
- Durable long-term storage
- Automatic horizontal scaling
- User / auth management

Prometheus's architecture



The data model

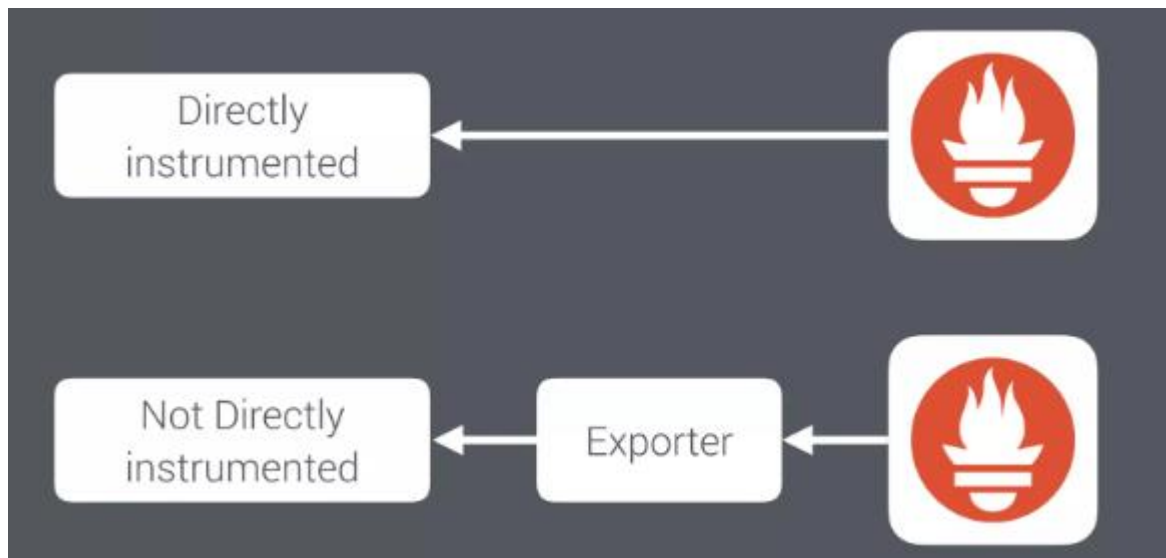
Every *time series* is uniquely identified by its *metric name* and a set of *key-value pairs*, also known as *labels*.

Notation:

```
<metric name>{<label name>=<label value>, ...}
```

Example: `api_http_request_total{method="POST", handler="/message"}`

How to get metrics ?



PromQL

- Non-SQL query language
- Better for metrics computation
- Only does reads

PrmQL – Operators

+	Addition	==	Equal
-	Substraction	!=	Not-equal
*	Mutiplication	>	Greater than
/	Division	<	Less then
%	Modulo	>=	Greater or equal
^	Exponentiation	<=	Less or equal
and	Intersection	or	Union
unless	complement		

PromQL – Aggregation operators

sum	stddev	bottomk
min	stdvar	topk
max	count	quantile
avg	count_values	

Alerting?

- Symptom-based alerting (be proactive)
- Prevent alert fatigue
 - ✓ Use ticketing systems (Avoid email spam)
 - ✓ Warning are tasks like new feature
- Provide runbooks
 - ✓ Keep them concise
 - ✓ Explanation, hints, links
 - ✓ Dynamic – include recent observations
- Practice outages

Module 2: Grafana

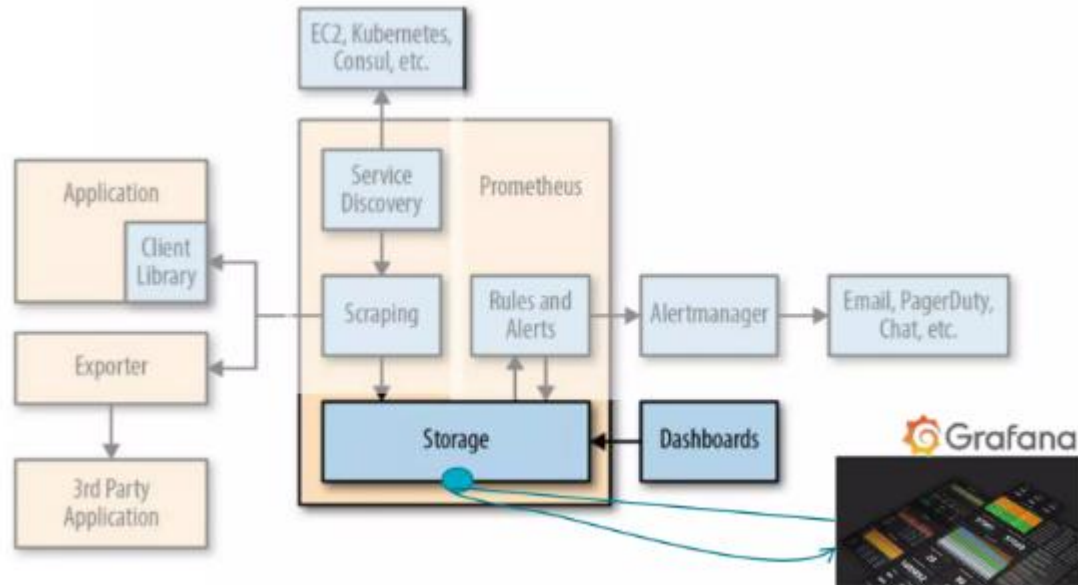


What is Grafana?

- A generic open-source dashboard product
 - ✓ Supporting many types of data sources
- Queries data sources periodically
 - ✓ Does not store any data
 - ✓ Refreshes visualizations
 - ✓ Evaluates alert conditions and triggers alerts/sends notifications
- Extensive library of pre-built dashboards
- Supports user authentication and authorization and multi-tenancy
- Online playground

Module 2: Grafana Introduction

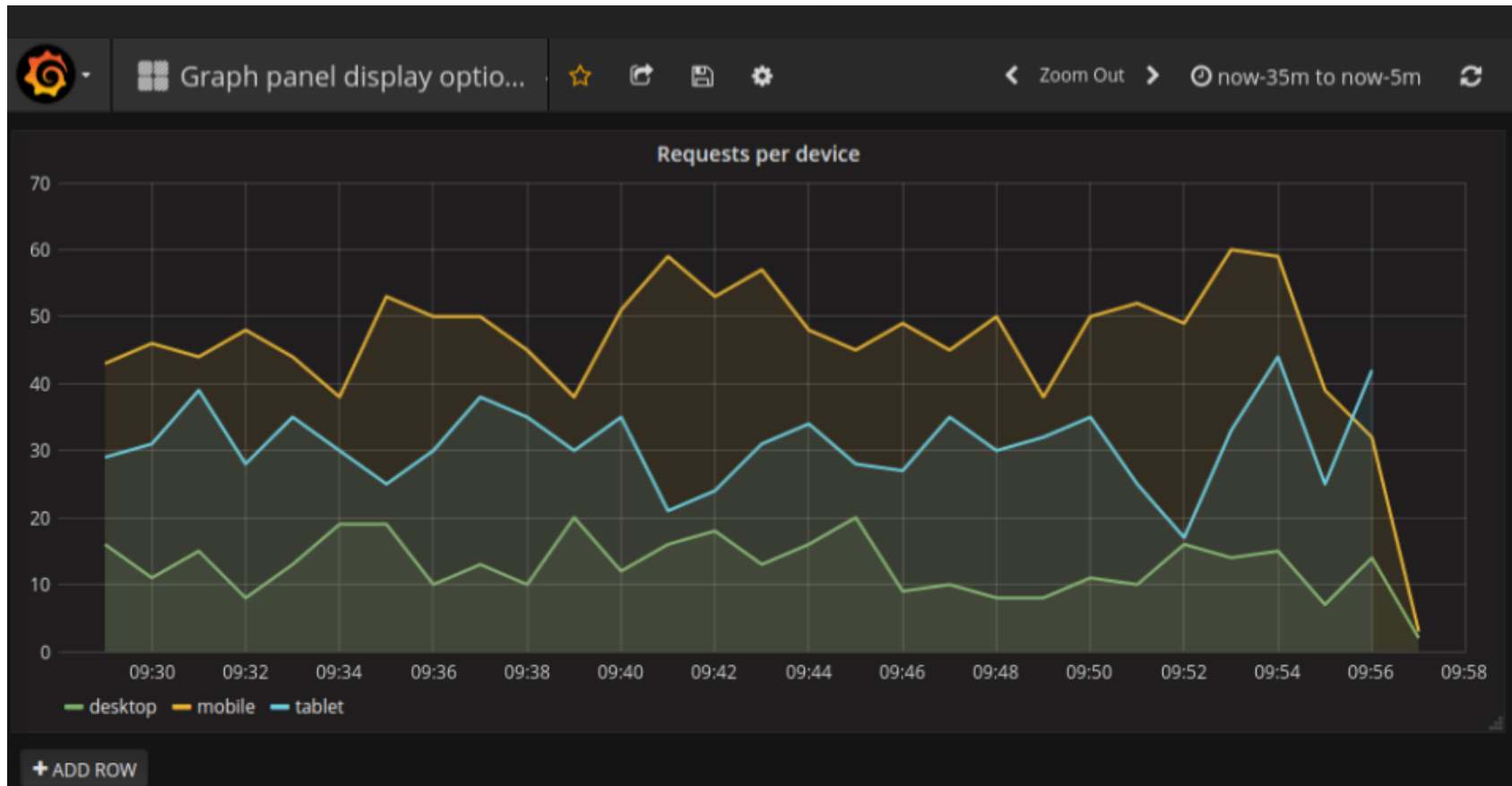
Grafana integrates with Prometheus



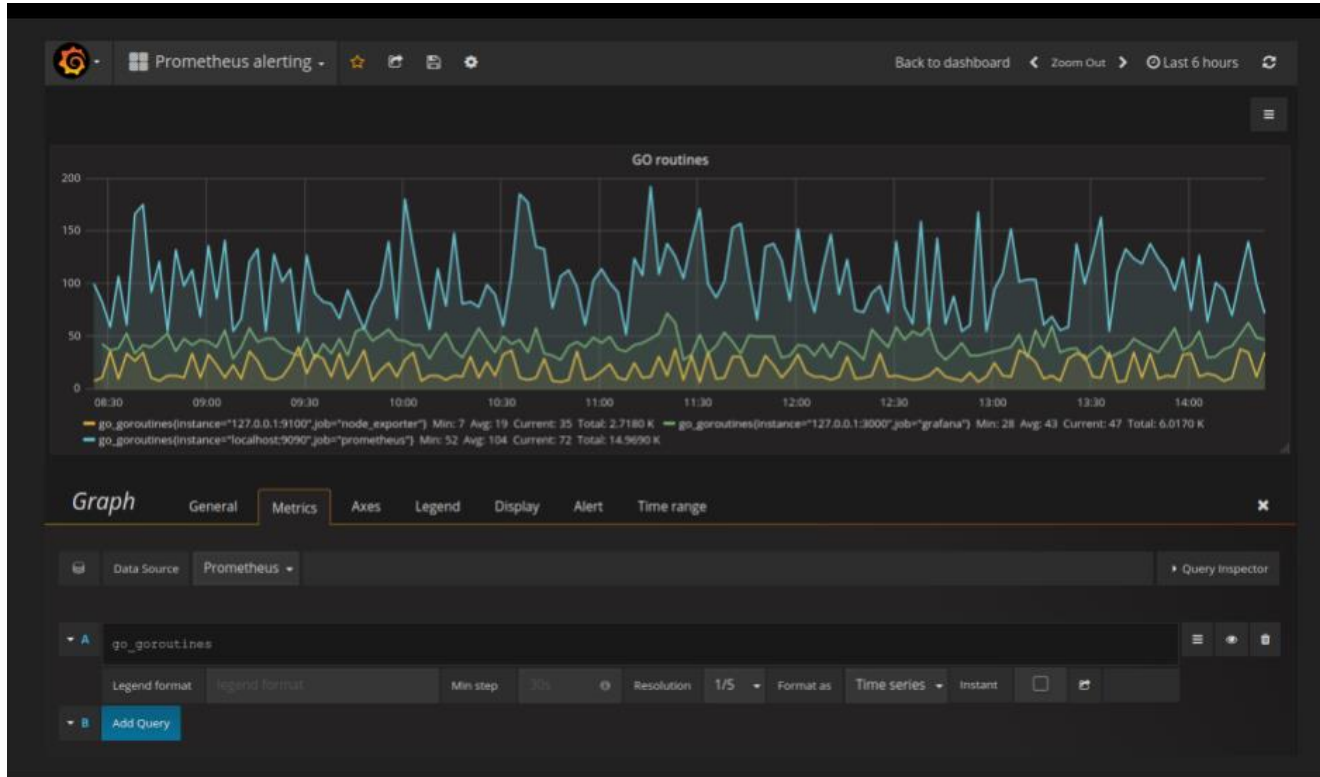
Module 2: Grafana Introduction



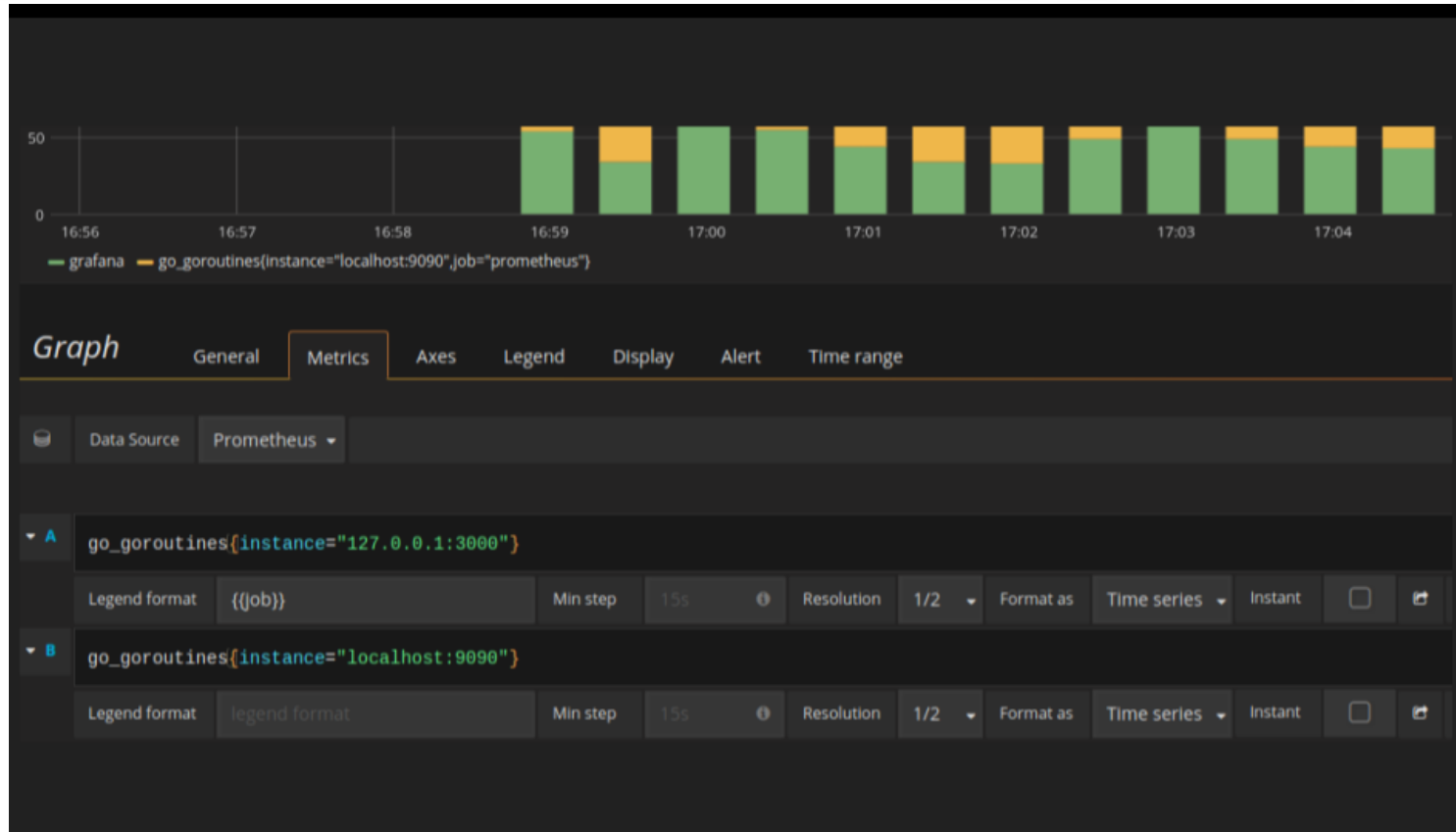
Module 2: Grafna Introduction



Module 2: Grafana Introduction



Module 2: Grafana Introduction



Hope is NOT
a strategy



Thank you

